Sudhanshu Rai

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EDUCATION

National Institute of Technology, Agartala — **Aug 2024 – May 2028**
Bachelor of Technology in Computer Science and Engineering | ***CGPA: 9.35/10*** — *Agartala, Tripura*

SRJSIC Ramghat Koilaswan Khurd, Deoria — **Jun 2022 – Apr 2023**
Senior Secondary (Class XII) | ***Percentage: 88.88%*** — *Kushinagar, Uttar Pradesh*

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, JavaScript, TypeScript, SQL
Web & UI: HTML5, CSS3, Responsive UI, React.js, React Router, Vite, Tailwind CSS
Backend & APIs: Node.js, Express.js, RESTful APIs, JWT Authentication, Clerk, bcrypt, Postman
Databases & Tools: MongoDB, Mongoose, MySQL, Git, GitHub, Vercel, Render, Stripe, Mailjet
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Game of Codes — Algorithmic Hackathon — **Mar 2026**
Algorithmic Programmer — Finalist — *NIT Agartala, Tripura*

- Engineered optimized solutions in **C++** for advanced algorithmic problems spanning graph theory, segment trees, and dynamic programming under strict $\mathcal{O}(N \log N)$ time constraints.
- Placed **2nd** in the offline finals among **100+ participating teams** by decoding complex problem constraints and selecting optimal data structures.

Smart India Hackathon (SIH 2025) — **Nov 2025**
Full Stack Developer — Team Member — *NIT Agartala, Tripura*

- Engineered a distributed backend architecture for real-time tourist tracking, handling high-throughput data streams for **1,000+ simulated concurrent users** using **Node.js** and **Express.js**.
- Spearheaded backend integration of modular **RESTful APIs** across a 6-member team using Git workflows; achieved **National Rank 7 among 500+ teams** nationwide (Top 1.4%).

PROJECTS

MediCare | Live Demo | GitHub — **Aug 2026 – Sep 2026**

- **Tech Stack:** React.js, JavaScript (ES6+), Tailwind CSS, Node.js, Express.js, MongoDB, Clerk, Stripe
- Architected a scalable full-stack healthcare platform with **Role-Based Access Control (RBAC)** across **3 user roles** (Patients, Doctors, Admins) and secure session handling via **Clerk**.
- Built **15+ modular REST APIs** in Node.js/Express for appointment workflows and added **MongoDB compound indexes** on queried fields to accelerate data retrieval.
- Integrated the **Stripe API** with webhook event listeners for end-to-end payment flows; deployed the frontend on **Vercel** using code-splitting and asset optimization.

Job Notification Platform | Live Demo | GitHub — **Jun 2026 – Jul 2026**

- **Tech Stack:** React.js, Vite, JavaScript, Node.js, Express.js, MongoDB, JWT, Mailjet
- Developed a centralized recruitment management platform with **JWT authentication**, protected client-side route guards, and dual-role access (Student/Admin) for announcements and task pipelines.
- Built scalable **CRUD REST endpoints** and integrated the **Mailjet API** for automated asynchronous batch email alerts to registered students; deployed the backend on **Render** and frontend on **Vercel**.

ACHIEVEMENTS

- **Competitive Programming:** Rated **Expert (1607)** on Codeforces (660+ solved), **Knight (2105)** on LeetCode (1030+ solved), and **3-Star (1641)** on CodeChef (200+ solved).
- **Global Contest Ranks:** Secured global ranks of **492** in Codeforces Round 1076, **787** in LeetCode Weekly Contest 505, and **127** in CodeChef Starters 219 (Div. 3) among 15,000+ competitors.
- **JEE Main 2024:** Achieved **top 1.87 percentile** nationwide out of **1.5 million** candidates.

LEADERSHIP & RESPONSIBILITY

Assistant General Secretary (AGS) — **Oct 2025 – Present**
Data Science and Artificial Intelligence Club — *NIT Agartala, Tripura*

- Directed **2 technical workshops** on AI/ML fundamentals and Python, upskilling **80+ student participants** through hands-on development.
- Orchestrated **campus-wide coding challenges** and hackathons, driving **100+ active student registrations** and promoting a competitive programming culture.